

ROSSMANN SALES FORECASTING

End-to-End Retail Sales & Customer Demand Forecasting

Project 6 • Final Professional Documentation

Submitted By

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GitHub: github.com/vsatish5621/Rossmann-Sales-Forecasting

Live App: rossmann-sales-forecasting-3ajs4nefpwuhytynunnisq.streamlit.app

1. Executive Summary

This project delivers an end-to-end Rossmann retail forecasting solution covering data preparation, business-focused EDA, feature engineering, Random Forest sales and customer models, a multivariate LSTM time-series model, MLflow tracking and inference, DVC data versioning, Power BI reporting and a deployed Streamlit application.

Metric	Final Result
Training records	1,017,209
Stores	1,115
Final test records	41,088
Sales RF	MAE 809.24 RMSE 1,218.28 R ² 0.8859
Customer RF	MAE 60.23 RMSE 91.12 R ² 0.9562
Multivariate LSTM	MAE 504,875.09 RMSE 669,092.44 R ² 0.9451
Forecast outputs	Sales + Customers for 41,088 test records

The solution is designed around the official business need: repeatable forecasts up to six weeks ahead using promotions, competition, school/state holidays, seasonality and locality/store context.

2. Official Requirement Coverage

The following matrix reflects the final implementation state.

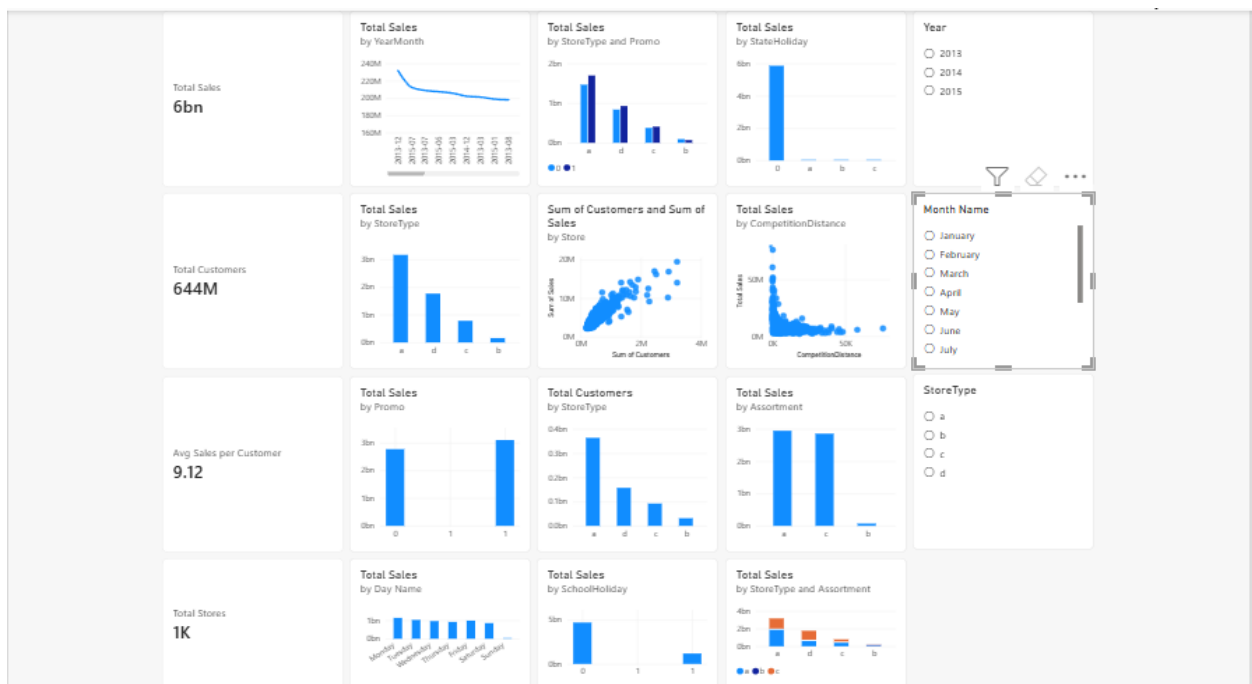
Requirement	Status	Final evidence
Task 1 EDA	DONE	Required questions, visualizations/tables and business insights.
Missing data	DONE	Pipeline imputation; structural Promo2 missingness documented.
Outlier handling	DOCUMENTED	Extreme observations reviewed as business observations; no unsupported claim of automated deletion.
Logging	DONE	Python logging implementation retained.
RF pipeline	DONE	sklearn preprocessing + Random Forest chronological validation.
Loss/evaluation	DONE	MSE training objective; MAE/RMSE/R ² business interpretation.
Feature importance	DONE	Top features documented; reliance $\neq$ causality.
Prediction interval	DONE	Approximate tree-dispersion interval; not calibrated.
Serialization	DONE	Timestamped RF/customer/LSTM/scaler artifacts.
LSTM	DONE	ADF, differencing decision, ACF/PACF, sliding windows, scaling and regression.
MLflow	DONE	Inference + registry Version 1 and Version 2 evidence.
Web interface	DONE	Manual + CSV batch, Sales/Customers, charts, CSV download.
Hosting	DONE	Streamlit Community Cloud live application.
GitHub/DVC	DONE	Five notebooks, app, DVC Version 1/2 evidence.
Final deliverables	DONE	PDF, 25-slide PPT, evidence and final predictions.

3. Data Understanding & EDA

The project uses 1,017,209 historical store-day records, a 1,115-store master table, 41,088 future test records and a 41,088-row submission template. Data quality checks found no duplicate rows. Store-level structural missingness is concentrated in competition and Promo2 fields.

Key business findings

- Sales–Customers correlation is 0.8947.
- Promotion increases average Sales from 4,406.05 to 7,991.15 and average Customers from 517.82 to 820.10.
- Promotion share is 38.15% in train versus 39.58% in test.
- State holidays materially suppress demand, with recovery around holiday periods.
- Christmas shows strong seasonal behaviour; Dec 25 is near-zero while Dec 23 and Dec 30 are stronger.
- Sunday is a major operational trough because many stores are closed.
- Assortment b has the highest observed average Sales.
- CompetitionDistance has weak overall correlation with Sales (-0.0192), so locality is multidimensional.
- SchoolHoliday has a notable train/test distribution shift and should be monitored in deployment.



Power BI dashboard evidence

4. Machine Learning — Sales & Customer Forecasting

A chronological split was used: January 2013–June 2015 for training and July 2015 for validation. The Sales model uses 20 features covering store, calendar, promotion, holiday, assortment, competition and Promo2 context. Customers were excluded from the Sales model to avoid future-information leakage.

Sales Random Forest

RandomForestRegressor: 100 trees, max_depth 20, min_samples_leaf 2, random_state 42.

Model	MAE	RMSE	R ²
Sales RF	809.24	1,218.28	0.8859
Customer RF	60.23	91.12	0.9562

Feature importance is led by Open (0.508711), CompetitionDistance (0.092245), Promo (0.080344) and Store (0.070218). Open dominates because closed stores naturally generate zero Sales; this is model reliance, not causality.

Prediction uncertainty is estimated from individual-tree dispersion using an approximate mean $\pm$ 1.96 $\times$ standard-deviation interval, with the lower bound clipped at zero. It is not a calibrated confidence interval.

5. Deep Learning — LSTM Time Series

Daily Sales were aggregated into a 942-day time series from 2013-01-01 to 2015-07-31. The ADF statistic was -4.7616 with $p \approx 6.4 \times 10^{-5}$, so the stationary path was retained and first-order differencing was not required. Strong weekly ACF/PACF structure motivated a 28-day sliding window.

Multivariate LSTM

The improved model uses daily Sales, OpenStores and DayOfWeek. Inputs were scaled to (-1,1) using training-only scalers. Architecture: Input (28,3) → LSTM(64) → Dropout(0.2) → Dense(1). Validation results: MAE 504,875.09, RMSE 669,092.44 and R^2 0.9451.

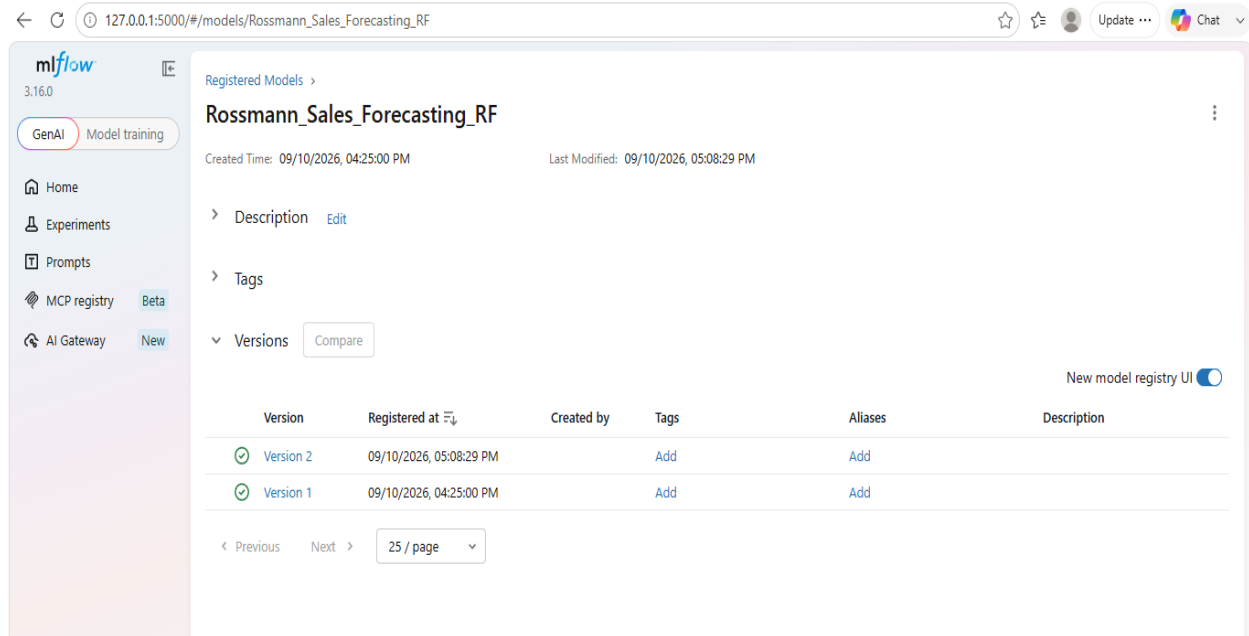
The multivariate model captured Sunday troughs substantially better than the univariate LSTM. RF and LSTM metrics are not directly comparable because RF operates at store-day granularity while the LSTM evaluation is on daily aggregated sales.

6. MLOps, Reproducibility & BI

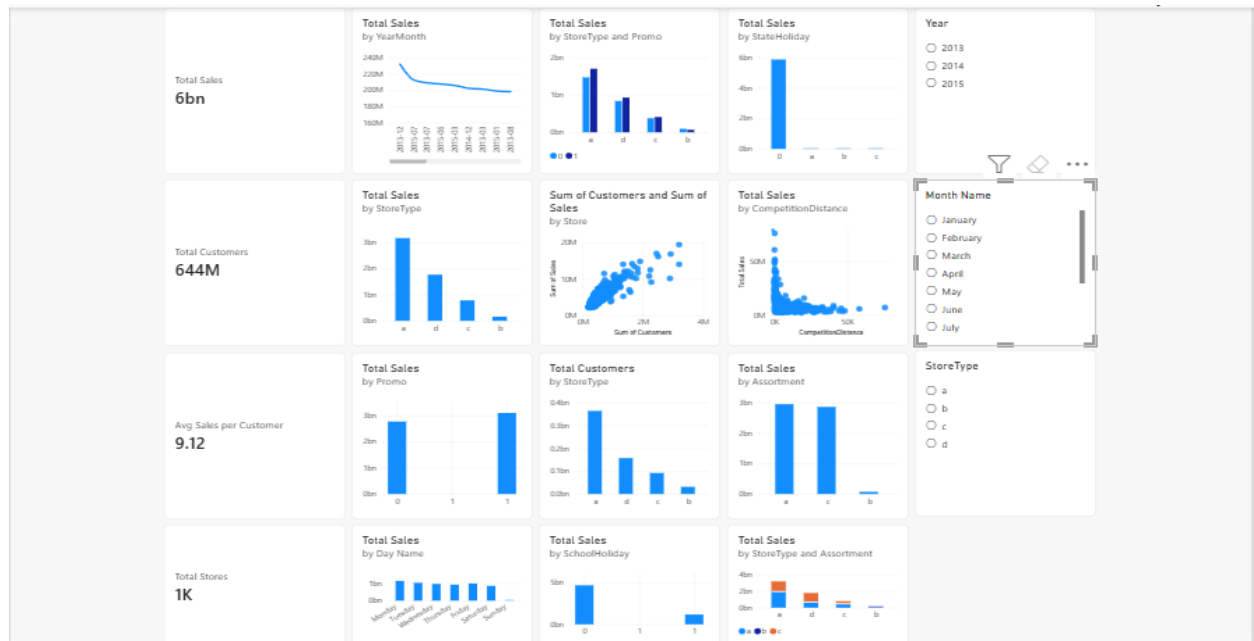
MLflow experiment tracking records parameters, metrics and model artifacts. MLflow inference reproduced 41,088 test predictions. Registry Version 1 and Version 2 are retained; Version 2 is a second registry release of the validated artifact, not a newly trained model.

DVC maintains two data versions: Version 1 commit 6fe968e for the original Rossmann datasets and Version 2 commit 5312ffc for the prepared training dataset. dvc diff confirmed that train.csv changed.

Power BI uses a star-schema model with FactSales, DimStore and DimDate, plus DAX measures for Total Sales, Total Customers, Average Sales per Customer and Total Stores. Year, Month and Store Type slicers support interactive business analysis.



MLflow registry evidence — Version 1 and Version 2



Power BI dashboard evidence

7. Web Application & Deployment

The final Streamlit application provides two workflows: manual single-store prediction and CSV batch prediction. The batch workflow accepts the model's required input columns, validates them, generates Predicted Sales and Predicted Customers, displays dedicated charts, and provides a downloadable CSV.

Live application

<https://rossmann-sales-forecasting-3ajs4nefpwuhytinxunnisq.streamlit.app>

The screenshot shows the 'Rossmann Sales Forecasting' web application interface. At the top, there's a title 'Rossmann Sales Forecasting' with a bar chart icon. Below it, a message states: 'Enter the store and business conditions below to predict Sales and Customer numbers.' A light blue box contains the text: 'The application uses trained Random Forest models for Sales and Customer prediction.' The main section is titled 'Store & Business Information' with a 'NEW' badge. It features several input fields: 'Store ID' (1), 'State Holiday' (No State Holiday), 'Competition Distance' (500.00), 'Prediction Date' (2015 / 07 / 31), 'School Holiday' (No), 'Competition Open Month' (0), 'Store Status' (Open), 'Store Type' (a), and 'Competition Open Year' (0). A 'Manage app' button is visible in the bottom right corner.

Streamlit manual prediction interface

Rossmann Sales Forecasting

Use the trained Random Forest models to forecast Sales and Customer numbers for Rossmann stores.

The application supports both manual single-store prediction and CSV-based batch prediction.

Choose Prediction Method

☐ Manual Prediction ☒ CSV Batch Prediction

CSV Upload & Batch Prediction

Upload a CSV containing the 20 model input features. The application will generate Sales and Customer predictions for every row.

[Download CSV Template](#)

Upload prediction CSV

[Upload](#) 200MB per file • CSV

Streamlit CSV batch prediction workflow

8. Final Outputs & Submission Access

The final combined prediction output contains 41,088 rows with Id, Predicted_Sales and Predicted_Customers. Timestamped model artifacts are retained separately, while the public GitHub repository contains the source notebooks, application and supporting files. Large model binaries are managed using the project repository's Git LFS/DVC strategy rather than being treated as ordinary text files.

Deliverable	Final status
Five organized notebooks	Included / ready
Final prediction CSV	41,088 rows
Power BI PBIX	Completed
Streamlit app	Live
Professional PDF	This document
Professional PPT	25 slides
MLflow evidence	Version 1 + Version 2
DVC evidence	Version 1 + Version 2

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9. Final Audit & Transparent Limitations

Final submission readiness: **READY**. The current implementation satisfies the core official requirements and the final evidence package contains the principal documentation and screenshots.

Transparent limitations

- The static store master does not provide a complete historical competitor-distance timeline for every store; the competitor NA/opening analysis is therefore documented within the available data constraints.
- Extreme demand observations were reviewed as legitimate business observations. The final workflow does not claim a separate automated outlier-removal transformation.
- The tree-dispersion prediction interval is approximate and not statistically calibrated.
- RF and LSTM metrics use different prediction granularities and should not be ranked directly.
- Feature importance is model reliance and should not be interpreted as causal impact.

Final checklist

- Official requirements reviewed
- Five notebooks retained
- EDA findings and evidence retained
- Random Forest Sales + Customer models retained
- LSTM stationarity/ACF/PACF/window/scaling evidence documented
- MLflow inference and registry evidence retained
- DVC multiple-version evidence retained
- Power BI dashboard evidence retained
- Streamlit CSV upload + Sales/Customers charts + CSV download verified
- GitHub and live app links included
- Final 25-slide PPT updated
- Final PDF and DOCX updated

Prepared for final submission — 15 September 2026